



Algebra 1 Honors

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 8–9 · FULL YEAR · HONORS

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in lesson order, examples, and activities from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Algebra 1 Honors introduces scholars to abstract mathematical thinking with the added rigor of an honors course: a capacity to see patterns, represent them symbolically, and reason about what must be mathematically true. This classically-aligned course engages scholars in rigorous problem-solving across five areas — operations with polynomials and radicals, the behavior and modeling power of functions, solving equations and systems, building and analyzing functions, and interpreting data. VR, AI, and spatial-computing enhancements build logical reasoning, conceptual fluency, and real-world application. Throughout, scholars develop precision in symbolic work and care in understanding what algebraic structures reveal. Algebra 1 Honors satisfies the Algebra 1 graduation requirement.

THE YEAR AT A GLANCE

1 Polynomials, Radicals, and Exponents (area 1)

Performing operations with polynomials and radicals, and extending the Laws of Exponents to include rational exponents.

3 Equations and Systems (area 3)

Solving quadratic equations in one variable and systems of linear equations and inequalities in two variables.

5 Data Analysis (area 5)

Representing and interpreting categorical and numerical data with one and two variables.

2 Linear, Quadratic, and Exponential Functions (area 2)

Extending functions to model and analyze real-world relationships.

4 Building and Analyzing Functions (area 4)

Building functions, identifying their key features, and representing them in various ways — including combining functions.

The Honors difference. Greater rigor throughout — increased text and task complexity, combining functions, and a deeper, more abstract treatment of structure and reasoning. This is direct preparation for Geometry Honors and the honors mathematics sequence.

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- VR, AI, and spatial-computing tools make patterns and function behavior visible; quizzes and tests close each unit.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A graphing or scientific calculator, used throughout the course.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.